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VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

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SUBJECT CODE : BMAT202L

SUBJECT TITLE : PROBABILITY AND STATISTICS

LAB SLOT : L47+L48

SEMESTER : Winter Semester 2025-2026

Experiment No. – 6

Chi-Square Test

1. The following table gives the number of fatal road accidents that occurred during the 7 days of a week. Write down the R programming code to test whether the accidents are uniformly distributed over the week at 95 % level of confidence.

Day :	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Number :	15	14	16	12	11	14	12

Mathematics

Aim

To write an R program to test whether the number of fatal road accidents is uniformly distributed over the 7 days of the week using a Chi-square goodness-of-fit test at 95% confidence level.

KEY WORDS

obs = Vector containing observed frequencies (number of accidents each day)
exp = Vector containing expected frequencies under uniform distribution
chisq.test() = Function used to perform Chi-square goodness-of-fit test
p = Vector of probabilities (equal for uniform distribution)
result = Variable used to store test output
Chi-square statistic = Measure of deviation between observed and expected values
p-value = Probability used for hypothesis testing

R Code

main.r Share Run Output Clear

```
obs=c(15,14,16,12,11,14,12)
exp=rep(sum(obs)/length(obs),length(obs))
result=chisq.test(obs,p=rep(1/7,7))
result
```

```
    Chi-squared test for given probabilities

data:  obs
X-squared = 1.4681, df = 6, p-value = 0.9616


=== Code Execution Successful ===
```

Conclusion

From the R output of the Chi-square goodness-of-fit test:

- Chi-square statistic = 1.4681
- p-value = 0.9616

Since the p-value is greater than 0.05 (95% confidence level), we fail to reject the null hypothesis. Thus, there is no significant difference between observed and expected frequencies, and the accidents can be considered to be uniformly distributed over the days of the week.

2. A total number of 3759 individuals were interviewed according to gender and decision in a public opinion survey on a political proposal with the results as in the following table. Write down the R programming code to test the hypothesis that there is no association between gender and attitude 5 % level of significance.

	Decision		
	Favoured	**Opposed**	**Undecided**
Male	1154	475	243
Female	1103	442	342

Aim

To write an R program to test whether there is no association between gender and attitude using a Chi-square test of independence at 5% level of significance.

KEY WORDS

data = Matrix representing contingency table (gender vs decision)
matrix() = Function used to create matrix
chisq.test() = Function used to perform Chi-square test of independence
nrow = Number of rows in matrix
byrow = TRUE fills matrix row-wise
result = Variable used to store test output
Chi-square statistic = Measure of association between variables
p-value = Probability used for hypothesis testing

R Code

main.r

```
data=matrix(c(1154,475,243,
              1103,442,342),
              nrow=2,byrow=TRUE)
result=chisq.test(data)
result
```

Output

```
	Pearson's Chi-squared test

data:  data
X-squared = 19.034, df = 2, p-value = 7.358e-05


=== Code Execution Successful ===
```

Conclusion

From the R output of the Chi-square test of independence:

- Chi-square statistic = 19.034
- p-value = 7.358×10^{-5}

Since the p-value is less than 0.05 (5% level of significance), we reject the null hypothesis.
Thus, there is a significant association between gender and attitude towards the political proposal.